Milestone II - VAERS insights
Applying Supervised and Unsupervised Models to Predict Fatalities

(by Leon Gutierrez, Tung Nguyen, and James Isioma)

Introduction

Prior to COVID-19, long term studies of vaccine reactions and fatalities were standard. Yet COVID-19 vaccines were approved under "Emergency Use Authorization" with limited exploration of possible fatal effects. Despite the Freedom of Information Act, pharmaceutical studies remain incomplete. This effort will analyze public data from VAERS and health data from CDC sites using supervised and unsupervised algorithms. We hope to find predictive factors and hidden trends tied to vaccine related fatalities, addressing incomplete records, sample imbalances, and biases. Challenges: However, VAERS depends on self-reported accounts (patients, hospitals, doctors, etc), causing differences in data quality. Because of this, many incidents are not reported. Related Work: "Profiling COVID-19 Vaccine Adverse Events" mapped VAERS data to identify adverse reactions in different COVID-19 vaccines. We build on that study by analyzing 2021-2022 data, using supervised and unsupervised models, and centering on fatalities. Our approach will addresses incomplete records, sample imbalances, and biases that can affect critical risk factor discovery. To address potential underreporting or overrepresentation in the VAERS data, we will explore weighting methods that account for demographic and reporting discrepancies.

Dataset and Data Preprocessing

Main Dataset: We will use 2021 and 2022 VAERS data as the primary database for the project, as these are the years with the most COVID-19 fatalities. This dataset contains self-reported information about adverse reactions following vaccination. The data is provided in three files: demographic, symptoms, and vaccine type.

Secondary Datasets: We will incorporate two additional sources from the CDC databases. The first is the Wonder Interactive Query System (all-cause mortality rates by state and gender). The second is the CDC Atlas Database (diabetes rates by states). Preliminary research suggests metabolic disorders may increase the risk of severe vaccine reactions, making state level diabetes rates a potentially informative feature. Currently exploring to add heart conditions.

Data Cleaning: Data will be pre-processed by joining the datasets, addressing missing values, and standardizing key variables. Our team will then engineer features such as top_comorbidities, time_since_vaccination, top_symtoms, etc. Due to the complexity of self-reported healthcare data, missing values, and inconsistent symptom/comorbidity reporting, careful data cleaning and feature engineering will be critical. These steps will help ensure that the data is both accurate and predictive.

Similar Databases: It is important to note that partial information from VAERS has been posted on Kaggle. Two sources with similar information were published. However in one case (Kaggle Dataset 1), the information is a small subset of what we will use, and in the other (Kaggle Dataset 2), the data includes information beyond the COVID-19 vaccine period. Neither of these suggests any target.

Supervised Learning

Objective: Predict the likelihood of fatal outcomes post COVID-19 vaccination (target in dataset: DIED).

Model and dataset: Logistic regression will be used as a baseline model for its simplicity and interpretability, enabling initial insights into key predictors of fatal outcomes.

Dataset: We will primarily use VAERS demographic dataset in addition to the clusters obtained from unsupervised models. We plan to address the class imbalance (normal in rare events like fatalities) by using resampling techniques and class-weight adjustments, while also implementing systematic methods to handle missing fields (imputation or exclusion criteria).

Advanced Models: While we will start with logistic regression, and XGBoost, we remain open to other algorithms if initial experiments indicate a better fit."

Evaluation Metrics: Precision, recall and AUC-ROC metrics. The team will pay particular attention to the challenge of rare-events.

Visualizations: We will present confusion matrices, ROC curves, and key feature plots to show how each model predicts fatalities. We also provide chart to show how our fine-tuning optimization arrived to our proposed estimates

Unsupervised Learning

Objective: Clustering adverse event patterns to potentially uncover hidden structures and associations in the data that could allow for a better Supervised model. We intend to assess whether certain clusters correlate with higher fatality rates or specific demographic factors.

Dataset: Using VAERS symptoms and demographic datasets (comorbidities only), we will extract four text based all four fields and comorbidities, processed using TF-IDF or word embeddings for representation. Normalize numerical creatures and integrate demographic variables as needed

Models: We intend to develop cluster labels or first principal component as additional feature in the supervised evaluation." (K-means, Hierarchical Clustering, DBSCAN), using dimensionality reduction methods such as PCA or t-SNE for better visualization and to handle high-dimensional data.

Evaluation Metrics: We plan to measure cluster performance with silhouette scores and the Calinski-Harabasz index

Visualizations: We will plot clusters, highlight heatmaps, and use choropleths to find possible spatial patterns.

Team Planning

Activities Assignment by Week					
Jan/18 - Jan/24	**Jan25 - Jan/31**	**Feb/1 - Feb/7**	**Feb/8 - Feb/14**	**Feb/15 - Feb/21**	**Feb/28 - Mar/3**
Data acquisition, integration, cleaning, sampling - **Leon** TF-IDF/Embeddings - **James** Code Templates for Clustering - **Tung**	Feature Engineering Proposal - **Leon** Feature Engineering construct - **James and Tung**	Unsupervised Learning Clusters and Dimentionality reduction- **James & Tung** Supervised Learning and Initial Model Evaluation - **Leon**	Cluster evaluation in Supervised model and Feature importance - **Leon** Hyper parameter tunning and Sensitivity Analysis- **James** Failure Analysis and Failyre/error analysis - **Tung**	Documentation draft - **All** (each one to document their section)	Finalize Report and Submission

Logistics and Project Management lead	Leon
Great Lakes expert and hyperparameter fine tunning lead	James
Group Video and clustering lead	Tung

References and Related Works

1. U.S. Department of Health and Human Services (HHS): VAERS Data Sets – Provides raw VAERS datasets, including adverse events, vaccines, and patient demographics.
2. CDC WONDER dataset: Wide-ranging Online Data for Epidemiologic Research – Provides a comprehensive query system for analyzing mortality data by gender, state, and other parameters.
3. CDC Atlas Database: Diabetes Surveillance System – Provides state level diabetes data, including trends, historical data, and risk factors.
4. Kaggle Dataset 1: COVID-19 Vaccine Adverse Reactions – Includes a subset of VAERS data from 2020-2024, focusing on COVID-19 vaccines. Limited fields and relatively low download frequency. Not a Kaggle Competition and no suggested goal.
5. Kaggle Dataset 2: COVID-19 VAERS Dataset – Contains partial VAERS data from January to November 2021, focusing on adverse reactions to COVID-19 vaccines. Limited scope and low user downloads. Not a Kaggle Competition and no suggested goal.
6. **RELATED WORK**: "Profiling COVID-19 Vaccine Adverse Events by Statistical and Ontology-Based Analysis" This study analyzes VAERS data to identify adverse events associated with different COVID-19 vaccines.